

Chapitre 10

Git Partie 2

Plan

1. Vue d'ensemble
2. État courant
3. `git add`
4. `git commit`
5. Undo

Vue d'ensemble : définitions

Base Tree

État de départ (sans les modifications)

Working Tree

État courant (avec les modifications) = Base Tree + modifications

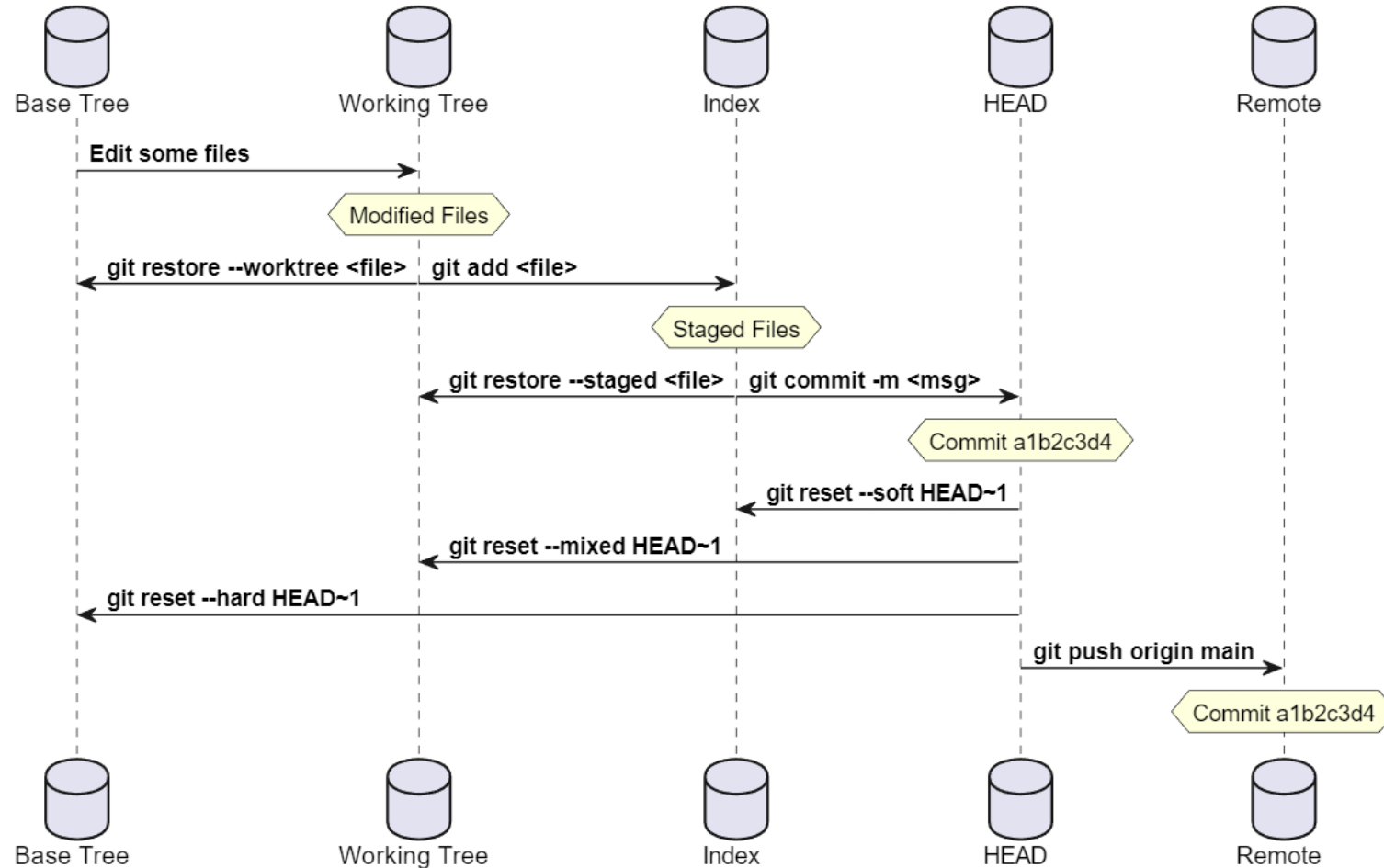
Index / Staging Area

Zone pour préparer le prochain commit

HEAD

Dernier commit de la branche actuelle

Vue d'ensemble



État courant

git status

repo

```
On branch main
Your branch is up to date with 'origin/main'.
```

index

```
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
       modified:   Le_Callennec_Benoit.txt
```

Working
Tree

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
       modified:   Le_Callennec_Benoit_2.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
       Le_Callennec_Benoit_3.txt
```

État courant

git diff différences entre Working tree et index

git diff --staged différences entre index et HEAD

```
diff --git a/Le_Callennec_Benoit_2.txt b/Le_Callennec_Benoit_2.txt
index d93666c..d1cf665 100644
--- a/Le_Callennec_Benoit_2.txt
+++ b/Le_Callennec_Benoit_2.txt
@@ -3,3 +3,4 @@ Version 0.2: parallel commit.
  Version 0.3: parallel commit 2.
  Version 0.4: parallel commit 3.
  Version 0.5: parallel commit coming from main.
+Version 0.6: some modifications to mark it as modified.
```

État courant

git log tous les commits

```
commit 7c37a406966c5e0401c8d829fb01d34492b89fd7 (HEAD -> main)
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sat Dec 2 19:02:48 2023 +0100

    DONE Commit for testing log command

commit 4681cfb6898c692b136e178011429f0307f3114f (origin/main, origin/HEAD)
Merge: 1358c5c 40818e5
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sun Nov 26 20:35:34 2023 +0100

    Merge branch 'BLC_MergeCommit'

commit 1358c5c48658575abb476d78174955c6f9341c19
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sun Nov 26 20:26:59 2023 +0100

    DONE A commit from main.

commit 40818e523bd9a605a9d9ae91fc72366df457a182
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sun Nov 26 20:25:58 2023 +0100

    DONE Some important features here.
```

État courant

git log -3 les 3 derniers commits

```
commit 7c37a406966c5e0401c8d829fb01d34492b89fd7 (HEAD -> main)
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sat Dec 2 19:02:48 2023 +0100

    DONE Commit for testing log command

commit 4681cfb6898c692b136e178011429f0307f3114f (origin/main, origin/HEAD)
Merge: 1358c5c 40818e5
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sun Nov 26 20:35:34 2023 +0100

    Merge branch 'BLC_MergeCommit'

commit 1358c5c48658575abb476d78174955c6f9341c19
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sun Nov 26 20:26:59 2023 +0100

    DONE A commit from main.
```

État courant

git log origin..HEAD les commits entre origin et HEAD

```
commit 7c37a406966c5e0401c8d829fb01d34492b89fd7 (HEAD -> main)
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sat Dec 2 19:02:48 2023 +0100

    DONE Commit for testing log command
```

État courant

git log --patch les commits avec les patches

```
commit 7c37a406966c5e0401c8d829fb01d34492b89fd7 (HEAD -> main)
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sat Dec 2 19:02:48 2023 +0100

    DONE Commit for testing log command

diff --git a/Le_Callennec_Benoit.txt b/Le_Callennec_Benoit.txt
index df21b7d..57e43bc 100644
--- a/Le_Callennec_Benoit.txt
+++ b/Le_Callennec_Benoit.txt
@@ -1,5 +1,4 @@
 Author: Benoit Le Callennec
 Version 0.1: first edit to this file.
 Version 0.2: second edit to this file to generate a merge commit.
-
-
\ No newline at end of file
+Version 0.3: some modifications to mark it as modified.

commit 4681cfb6898c692b136e178011429f0307f3114f (origin/main, origin/HEAD)
Merge: 1358c5c 40818e5
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date: Sun Nov 26 20:35:34 2023 +0100

    Merge branch 'BLC_MergeCommit'
```

État courant

git log <file> **les commits modifiant <file>**

```
commit 40818e523bd9a605a9d9ae91fc72366df457a182
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date:   Sun Nov 26 20:25:58 2023 +0100

    DONE Some important features here.

diff --git a/Le_Callennec_Benoit.txt b/Le_Callennec_Benoit.txt
index 0a64225..df21b7d 100644
--- a/Le_Callennec_Benoit.txt
+++ b/Le_Callennec_Benoit.txt
@@ -1,4 +1,5 @@
  Author: Benoit Le Callennec
  Version 0.1: first edit to this file.
+Version 0.2: second edit to this file to generate a merge commit.

\ No newline at end of file

commit 58b069ab95a7bbda6face500aa2ee16ebb4f5d2f
Author: Benoit Le Callennec <benoit.lecallennec@he-arc.ch>
Date:   Mon Dec 5 14:52:11 2022 +0100

    DONE Edited author file.
```

État courant

git log -- <file> si <file> a été effacé par le passé

git add – part 2

git add

Que se passe-t-il si on modifie un fichier déjà ajouté dans l'index ?

git add – part 2

git add

On branch main

Your branch is ahead of 'origin/main' by 1 commit.

(use "git push" to publish your local commits)

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: Le_Callennec_Benoit_2.txt

Untracked files:

(use "git add <file>..." to include in what will be committed)

Le_Callennec_Benoit_3.txt

git add – part 2

git add

On branch main

Your branch is ahead of 'origin/main' by 1 commit.

(use "git push" to publish your local commits)

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: Le_Callennec_Benoit_2.txt

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: Le_Callennec_Benoit_2.txt

Untracked files:

(use "git add <file>..." to include in what will be committed)

Le_Callennec_Benoit_3.txt

git commit --verbose

git commit --verbose

```
# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch main
# Your branch is ahead of 'origin/main' by 1 commit.
#   (use "git push" to publish your local commits)
#
# Changes to be committed:
#   modified:   Le_Callennec_Benoit_2.txt
#
# Untracked files:
#   Le_Callennec_Benoit_3.txt
#
# ----- >8 -----
# Do not modify or remove the line above.
# Everything below it will be ignored.
diff --git a/Le_Callennec_Benoit_2.txt b/Le_Callennec_Benoit_2.txt
index d93666c..74a0b8c 100644
--- a/Le_Callennec_Benoit_2.txt
+++ b/Le_Callennec_Benoit_2.txt
@@ -3,3 +3,5 @@ Version 0.2: parallel commit.
    Version 0.3: parallel commit 2.
    Version 0.4: parallel commit 3.
    Version 0.5: parallel commit coming from main.
+Version 0.6: some modifications to mark it as modified.
+Version 0.7: some modifications while already in index.
```

```
git commit --all
```

[git commit --all](#)

```
git add -u
```

```
git commit ...
```

Ne commit pas les nouveaux fichiers

Dangereux car on peut commit des modifications par erreur

Undo

git restore --worktree <file>

Undo toutes les modifications courantes faites à <file>

Undo

`git restore --staged <file>`

Retire les modifications faites à <file> depuis l'index

Identifier un commit

Avec son hash

Avec sa branche → dernier commit sur cette branche

HEAD → le dernier commit

HEAD~1 → l'avant-dernier / pénultième commit

HEAD~2 → l'antépénultième

HEAD~3 → Le préantépénultième

Identifier plusieurs commits

`<from> .. <to>`

`HEAD~3 .. HEAD~1` → tous les commits entre le
préantépénultième et l'avant dernier.

`origin/main .. HEAD` → tous les commits à push sur
`origin/main`

Undo - Suite

`git reset --soft <commit>`

`git reset --mixed <commit>`

`git reset --hard <commit>`

Déplace **HEAD** sur le commit spécifié

- `--soft` : les modifications sont gardées dans l'index
- `--mixed` : les modifications sont gardées dans WorkTree
- `--hard` : les modifications sont perdues

Vue d'ensemble

